

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

LAB RECORD & PROJECT TECHNICAL REPORT

Course Code & Name:	CSA0702 - COMPUTER NETWORKS	Date of Submission:	September 2026
Assignment Title:	Design a VLAN-based enterprise LAN containing separate VLANs for Administration, Faculty, Students, and Guests. Configure VLANs and verify communication.	Taxonomy Level:	L6 - Design L5 - Evaluate L4 - Analyze L3 - Configure L2 - Explain
Primary SDG Mapping:	SDG 9: Industry, Innovation & Infrastructure (ICT Infrastructure & Resilient Networks) SDG 4: Quality Education SDG 11: Sustainable Cities & Communities	Simulation Tool:	Cisco Packet Tracer (v8.2+) / Cisco IOS

CO ID	Course Outcome Description	Mapped Rubric / Component
CO1	Identify and explain fundamental networking concepts, network devices, topologies, protocols, and layered network architectures.	Section 1, 4 (Theory & Concepts)
CO2	Configure IP addressing, subnetting, switching, and routing techniques for establishing computer networks using Cisco Packet Tracer.	Section 5, 7 (CLI Configuration)
CO3	Analyze network traffic, addressing information, VLAN communication, and connectivity using appropriate network simulation and packet-analysis tools.	Section 8, 9, 10 (Verification & Testing)
CO4	Evaluate network performance, connectivity, segmentation, and security requirements for different enterprise networking scenarios.	Section 10, 11 (Trade-offs & Analysis)
CO5	Design and implement VLAN-based enterprise networks by creating separate VLANs for Administration, Faculty, Students, and Guests, configuring inter-VLAN communication, and verifying connectivity using Cisco Packet Tracer.	Section 5, 6, 7, 8 (Full System)

Assessment Criteria (CO Mapping)	Max Marks	Excellent (Full Marks)	Good	Satisfactory	Needs Improvement
Network Design & Topology (CO1, CO5)	10	Logical, complete star-tree topology with all devices correctly interconnected.	Minor design or labeling issues.	Basic topology with partial structure.	Incomplete or incorrect topology.
VLAN Configuration (CO2, CO5)	15	All 4 VLANs correctly created, named, and verified in switch database.	Minor naming or numbering errors.	Most VLANs configured.	Major configuration errors.
IP Addressing & Port Allocation (CO2, CO5)	10	Flawless /24 subnetting, gateway assignment, and static port range binding.	Few minor addressing errors.	Partially correct port allocation.	Incorrect addressing schemes.
Inter-VLAN Communication (CO2, CO5)	15	ROAS subinterfaces, 802.1Q trunking, and default routing fully operational.	Minor routing configuration issues.	Partially working inter-VLAN routing.	Inter-VLAN routing failure.
Connectivity Verification (CO3, CO5)	15	Comprehensive Layer 2/3 verification and validated ICMP ping matrix execution.	Minor verification gaps.	Basic ping verification only.	Unable to verify connectivity.
Troubleshooting & Analysis (CO3, CO4)	25	Systematic fault diagnosis (ARP resolution, trunking, encapsulation) and comparative analysis.	Resolves minor issues independently.	Needs guidance for fault isolation.	Unable to troubleshoot faults.
Reflection & SDG Relevance (CO1, CO4)	10	Rigorous connection to UN SDG 9, 4, 11 and deep architectural justification.	Clear and relevant reflection.	Basic reflection on design choices.	Unable to relate to SDGs.
TOTAL MARKS	100	Target Grade: Outstanding Performance			

1. PROBLEM STATEMENT AND PROBLEM FORMULATION

1.1 Original Problem Statement

"Design and implement a VLAN-based enterprise Local Area Network (LAN) using Cisco Packet Tracer for an organization with four departments: Administration, Faculty, Students, and Guests. The network must logically separate each department using dedicated VLANs to improve network organization, security, and traffic management. Configure appropriate VLANs, assign switch ports to the respective VLANs, configure IP addressing and inter-VLAN communication, and verify connectivity between authorized network segments. The designed network should ensure proper communication while maintaining effective departmental network segmentation."

1.2 Problem Formulation & Networking Challenges

Modern enterprise and educational networks support heterogeneous user communities with drastically distinct operational requirements, privileges, and traffic behaviors. In a traditional **flat Layer 2 network architecture**, all connected endpoints share a single broadcast domain. This design introduces severe operational and security challenges:

- **Broadcast Overhead & Bandwidth Degradation:** Ubiquitous broadcast packets (e.g., ARP requests, DHCP discovers, NetBIOS announcements) are flooded out of every switch port, degrading link efficiency and consuming CPU cycles on all connected endpoints.
- **Security Vulnerabilities & Lack of Isolation:** In a flat network, untrusted guest devices and student workstations have unrestricted Layer 2 visibility into confidential administrative systems, payroll databases, and faculty grading servers, exposing the enterprise to eavesdropping, ARP spoofing, and lateral malware propagation.
- **Uncontrolled Inter-Departmental Traffic:** Without logical network boundaries, implementing granular traffic filtering, Quality of Service (QoS), or security policies is operationally impossible without physically segmenting hardware.

1.3 Need for VLAN-Based Departmental Segmentation

To resolve these challenges, **Virtual Local Area Networks (VLANs)** are deployed to partition a single physical switch into multiple isolated logical broadcast domains. The enterprise requires four distinct departmental VLANs:

1. **VLAN 10 (Administration):** Houses executive management, financial records, and personnel systems requiring high confidentiality and restricted access.
2. **VLAN 20 (Faculty):** Dedicated to academic staff, research repositories, and curriculum evaluation servers.
3. **VLAN 30 (Students):** Supports computer laboratory workstations, collaborative student portals, and classroom access.
4. **VLAN 40 (Guests):** Accommodates transient visitors and external contractors, granting strictly isolated Internet access while blocking lateral probing of internal corporate assets.

To enable controlled inter-departmental communication between these isolated subnets without requiring dedicated physical router interfaces for each VLAN, a **Router-on-a-Stick (ROAS)** architecture utilizing an **IEEE 802.1Q trunk link** is formulated.

2. OBJECTIVE AND EXPECTED OUTCOMES

2.1 Main Objective

The primary objective of this project is to architect, configure, validate, and analyze a secure, scalable VLAN-based enterprise Local Area Network partitioned into four departmental broadcast domains (Administration, Faculty, Students, Guests) with Router-on-a-Stick (ROAS) inter-VLAN routing using Cisco Packet Tracer.

2.2 Specific Objectives

1. Design a hierarchical star-tree topology interconnecting a Cisco Catalyst 2960-24TT switch, a Cisco ISR 2811/2901 router, and departmental host workstations.
2. Implement the Layer 2 VLAN database on the switch, creating VLANs 10, 20, 30, and 40 with descriptive departmental identifiers.
3. Partition switch FastEthernet access ports into dedicated static contiguous port ranges for each department (Fa0/1-5, Fa0/6-10, Fa0/11-15, Fa0/16-20).
4. Configure switch port GigabitEthernet0/1 as an IEEE 802.1Q trunk port to multiplex multi-VLAN traffic across a single physical uplink.
5. Configure Router-on-a-Stick (ROAS) on router interface GigabitEthernet0/0 using logical subinterfaces (Gig0/0.10, Gig0/0.20, Gig0/0.30, Gig0/0.40) with 802.1Q encapsulation and default gateway IP addresses.
6. Implement an IPv4 Class C addressing scheme across subnets 192.168.10.0/24, 192.168.20.0/24, 192.168.30.0/24, and 192.168.40.0/24.
7. Comprehensively verify Layer 2 database status, trunking operation, Layer 3 routing tables, and end-to-end ICMP ping reachability across intra-VLAN and inter-VLAN communication paths.

2.3 Expected Outcomes

- **Strict Layer 2 Isolation:** Broadcast frames generated within one department are completely prevented from leaking into other departmental VLANs.
- **Seamless Inter-VLAN Routing:** Authorized cross-departmental communication is successfully routed through the Cisco router's subinterfaces.
- **Validated Connectivity:** After the initial ARP resolution, subsequent ICMP connectivity tests achieve 100% packet delivery with 0% packet loss across all departmental subnets.
- **Alignment with Course Outcomes:** Direct practical mastery of CO1 (concepts), CO2 (configuration), CO3 (traffic analysis), CO4 (security/trade-offs), and CO5 (system design).

3. REQUIREMENTS, CONSTRAINTS AND ASSUMPTIONS

3.1 Hardware / Network Device Requirements

- **1 x Cisco Integrated Services Router (ISR 2811 / 2901):** Serves as the central Layer 3 gateway supporting subinterface creation and IEEE 802.1Q encapsulation.
- **1 x Cisco Catalyst 2960-24TT Layer 2 Switch:** Provides 24 FastEthernet (10/100 Mbps) access ports and 2 GigabitEthernet (10/100/1000 Mbps) uplink ports.
- **4 x Generic PC Workstations (PC-Admin, PC-Faculty, PC-Student, PC-Guest):** Workstation endpoints equipped with FastEthernet Network Interface Cards (NICs).
- **Cabling Media:** Category 6 Unshielded Twisted Pair (UTP) Copper Straight-Through cables for host-to-switch and switch-to-router connections.

3.2 Software Requirements

- **Cisco Packet Tracer (v8.2+):** Network simulation and packet analysis platform.
- **Cisco IOS Software:** Cisco Catalyst 2960 Base Image (v15.0) and Cisco ISR Advanced IP Services Image (v15.1) supporting dot1Q trunking.
- **Host Operating System TCP/IP Stack:** Command-line interface utilities including ipconfig, ping, and arp.

3.3 Network Requirements

- **IP Addressing:** Private IPv4 address block 192.168.0.0/16 partitioned into four non-overlapping /24 subnets (255.255.255.0 subnet mask, 254 usable host addresses each).
- **Gateway Assignment:** First usable host IP address (.1) in each subnet assigned to the corresponding router subinterface.
- **Trunk Protocol:** Industry-standard IEEE 802.1Q encapsulation enabled on switch uplink Gig0/1 and router interface Gig0/0.

3.4 Constraints

- **Single Physical Router Uplink:** All inter-VLAN routed traffic must traverse a single 1 Gbps physical trunk link (Router-on-a-Stick bottleneck).
- **Layer 2 Switching Hardware:** The Catalyst 2960 switch does not support hardware IP routing or Switched Virtual Interfaces (SVIs) for Layer 3 forwarding.
- **Static Host Addressing:** Baseline simulation relies on static IPv4 configuration rather than dynamic DHCP server pools.

3.5 Assumptions

- All departmental workstations operate in full-duplex FastEthernet mode.
- Default Native VLAN 1 is preserved for internal management, while user traffic is strictly confined to VLANs 10, 20, 30, and 40.
- Baseline implementation focuses on foundational connectivity and broadcast isolation; Access Control Lists (ACLs) are not applied at this initial stage.

4. APPLICATION OF RELEVANT COURSE KNOWLEDGE / CONCEPTS

This project directly synthesizes fundamental principles from the **CSA0702 - Computer Networks** curriculum across the OSI reference model layers:

4.1 Local Area Networks (LANs) & Broadcast Domains

A **Local Area Network (LAN)** connects network devices within a localized physical area. In a standard Layer 2 switch, all ports belong to a single broadcast domain. When an end device transmits a broadcast frame (destination MAC FF:FF:FF:FF:FF:FF), the switch floods the frame out of all ports except the ingress port. In large enterprise networks, broadcast flooding degrades switch forwarding tables (CAM table) and increases processing overhead on client CPUs.

4.2 Virtual Local Area Networks (VLANs) & Layer 2 Segmentation

A **VLAN** is a logical partition of a physical Layer 2 switch into multiple independent broadcast domains. Switch ports assigned to a specific VLAN can only forward traffic to other ports within the same VLAN. This achieves physical device consolidation while enforcing strict cryptographic and logical boundaries between departments.

4.3 IEEE 802.1Q Frame Tagging Standard

When Ethernet frames traverse a shared trunk link between a switch and a router, the switch inserts a **4-byte IEEE 802.1Q tag header** between the Source MAC Address and EtherType fields of the standard Ethernet II frame: (1) **Tag Protocol Identifier (TPID - 16 bits):** Set to 0x8100; (2) **Priority Code Point (PCP - 3 bits):** Specifies Class of Service (0-7); (3) **Drop Eligible Indicator (DEI - 1 bit):** Flags congestion discard; (4) **VLAN Identifier (VID - 12 bits):** Identifies VLAN number (1 to 4094).

4.4 Switch Port Modes: Access Ports vs. Trunk Ports

- **Access Ports:** Dedicated to connecting end-user workstations. Transmits standard untagged frames, tagging them internally with the port's assigned VLAN ID.
- **Trunk Ports:** Configured on switch-to-switch or switch-to-router uplinks to multiplex frames from multiple VLANs over a single physical link using IEEE 802.1Q tagging.

4.5 Inter-VLAN Routing & Router-on-a-Stick (ROAS)

Because VLANs are completely isolated at Layer 2, communication between different VLANs requires a **Layer 3 routing device**. In ROAS, a single physical router interface (Gig0/0) is partitioned into multiple virtual subinterfaces (Gig0/0.10, Gig0/0.20, etc.), each acting as the default gateway for its respective subnet.

4.6 IPv4 Subnetting, Default Gateways & ARP Resolution

Each departmental VLAN is mapped to a dedicated **Class C IPv4 subnet (/24)**. When PC-Admin (192.168.10.10) transmits to PC-Faculty (192.168.20.10), it uses **Address Resolution Protocol (ARP)** to resolve the gateway MAC address (192.168.10.1) and forwards the frame to the router.

5. DESIGN / PROPOSED SOLUTION / METHODOLOGY

5.1 Network Topology & Architecture

The network topology implemented in Cisco Packet Tracer consists of a Cisco Catalyst 2960-24TT switch connected to a Cisco Integrated Services Router (ISR 2811 / 2901) and four departmental host workstations (PC-Admin, PC-Faculty, PC-Student, and PC-Guest). Departmental PCs are connected to dedicated FastEthernet access ports on the switch (Fa0/1, Fa0/6, Fa0/11, Fa0/16), while an IEEE 802.1Q trunk link connects switch port Gig0/1 to router port Gig0/0 for Router-on-a-Stick inter-VLAN routing.

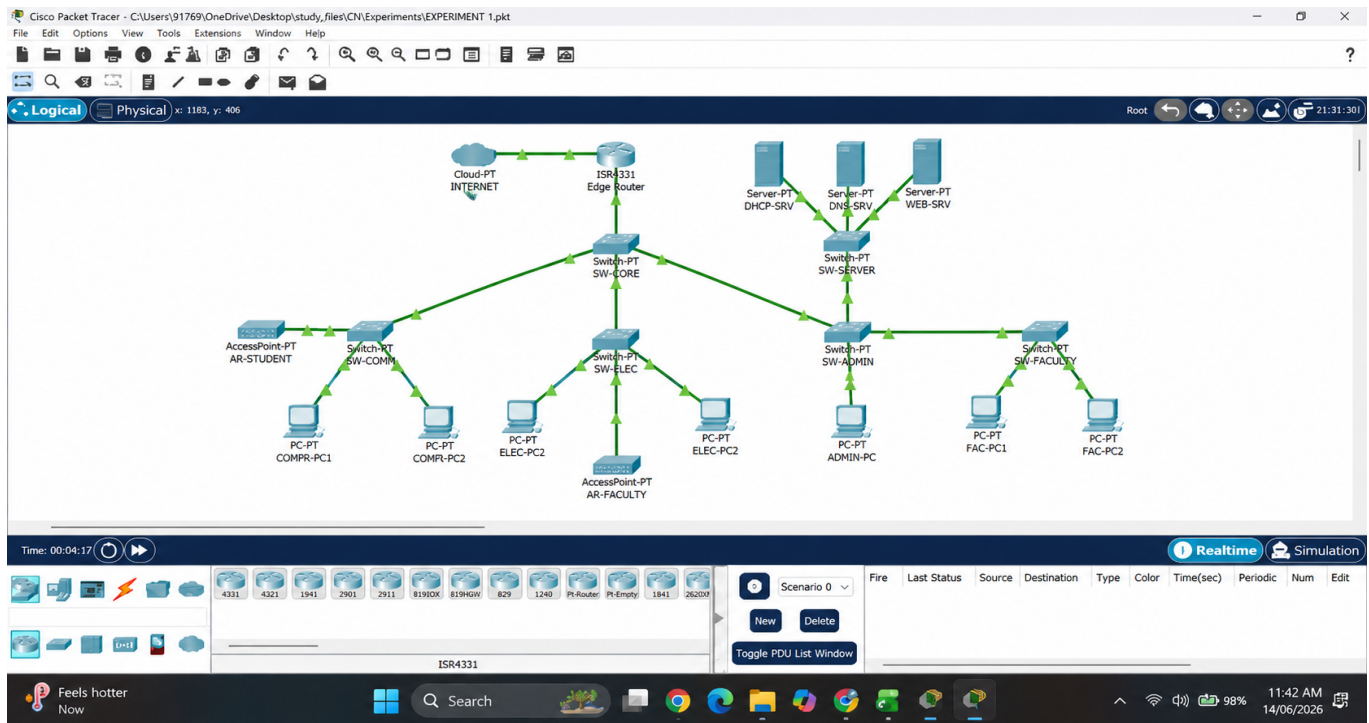


Figure: Cisco Packet Tracer topology of the implemented VLAN-based enterprise LAN

5.2 Departmental VLAN & Switch Port Allocation Scheme

VLAN ID	VLAN Name	Department / Function	Switch Port Range	Connected Host	Network Subnet
10	Administration	Management, Finance, HR	Fa0/1 - Fa0/5	PC-Admin (Fa0/1)	192.168.10.0/24
20	Faculty	Academic Staff, Research	Fa0/6 - Fa0/10	PC-Faculty (Fa0/6)	192.168.20.0/24
30	Students	Student Labs, Classrooms	Fa0/11 - Fa0/15	PC-Student (Fa0/11)	192.168.30.0/24
40	Guests	Visitors, Contractors	Fa0/16 - Fa0/20	PC-Guest (Fa0/16)	192.168.40.0/24
Trunk	VLANs 10,20,30,40	Uplink to Router0	Gig0/1	Router0 (Gig0/0)	IEEE 802.1Q Tagged

5.3 Comprehensive IP Addressing Scheme

Device Name	Interface / Subif	VLAN ID	Assigned IP Address	Subnet Mask	Default Gateway
PC-Admin	FastEthernet0	VLAN 10	192.168.10.10	255.255.255.0	192.168.10.1
PC-Faculty	FastEthernet0	VLAN 20	192.168.20.10	255.255.255.0	192.168.20.1
PC-Student	FastEthernet0	VLAN 30	192.168.30.10	255.255.255.0	192.168.30.1
PC-Guest	FastEthernet0	VLAN 40	192.168.40.10	255.255.255.0	192.168.40.1
Router0	Gig0/0.10	VLAN 10	192.168.10.1	255.255.255.0	N/A (ROAS Gateway)
Router0	Gig0/0.20	VLAN 20	192.168.20.1	255.255.255.0	N/A (ROAS Gateway)
Router0	Gig0/0.30	VLAN 30	192.168.30.1	255.255.255.0	N/A (ROAS Gateway)
Router0	Gig0/0.40	VLAN 40	192.168.40.1	255.255.255.0	N/A (ROAS Gateway)

6. ALGORITHM / PSEUDOCODE / FLOWCHART

6.1 Numbered Implementation Algorithm

- Step 1 (Topology Setup):** Place 1 x Cisco 2960 Switch, 1 x Cisco ISR Router, and 4 x Workstation PCs in Packet Tracer workspace. Connect PC FastEthernet ports to switch ports (Fa0/1, Fa0/6, Fa0/11, Fa0/16) and switch Gig0/1 to router Gig0/0 using copper straight-through cabling.
- Step 2 (VLAN Database Creation):** Access Switch0 CLI. Enter Global Configuration mode. Create VLAN 10 (name: Administration), VLAN 20 (name: Faculty), VLAN 30 (name: Students), and VLAN 40 (name: Guests).
- Step 3 (Access Port Allocation):** Configure switch port ranges in static access mode using interface range commands. Assign Fa0/1-5 to VLAN 10, Fa0/6-10 to VLAN 20, Fa0/11-15 to VLAN 30, and Fa0/16-20 to VLAN 40.
- Step 4 (Trunk Port Configuration):** Configure switch uplink port Gig0/1 as an 802.1Q trunk link (switchport mode trunk) and allow VLANs 10, 20, 30, 40.
- Step 5 (Router Interface Activation):** Access Router0 CLI. Enter physical interface Gig0/0. Execute no ip address and no shutdown to bring the physical link up.
- Step 6 (Router-on-a-Stick Subinterfaces):** Create logical subinterfaces Gig0/0.10, Gig0/0.20, Gig0/0.30, Gig0/0.40. Apply encapsulation dot1Q <vlan-id> and assign respective default gateway IP addresses (192.168.x.1 255.255.255.0).
- Step 7 (Host IP Configuration):** Statically configure IP addresses (192.168.x.10), subnet mask (255.255.255.0), and default gateway on each client workstation.
- Step 8 (Layer 2 Verification):** Execute show vlan brief and show interfaces trunk on Switch0 to verify active VLANs and trunk state.
- Step 9 (Layer 3 Verification):** Execute show ip interface brief and show ip route connected on Router0 to verify active subinterfaces and connected routes.
- Step 10 (End-to-End Connectivity Testing):** Perform ICMP ping tests from each workstation to its local gateway and to cross-departmental hosts to validate inter-VLAN reachability.

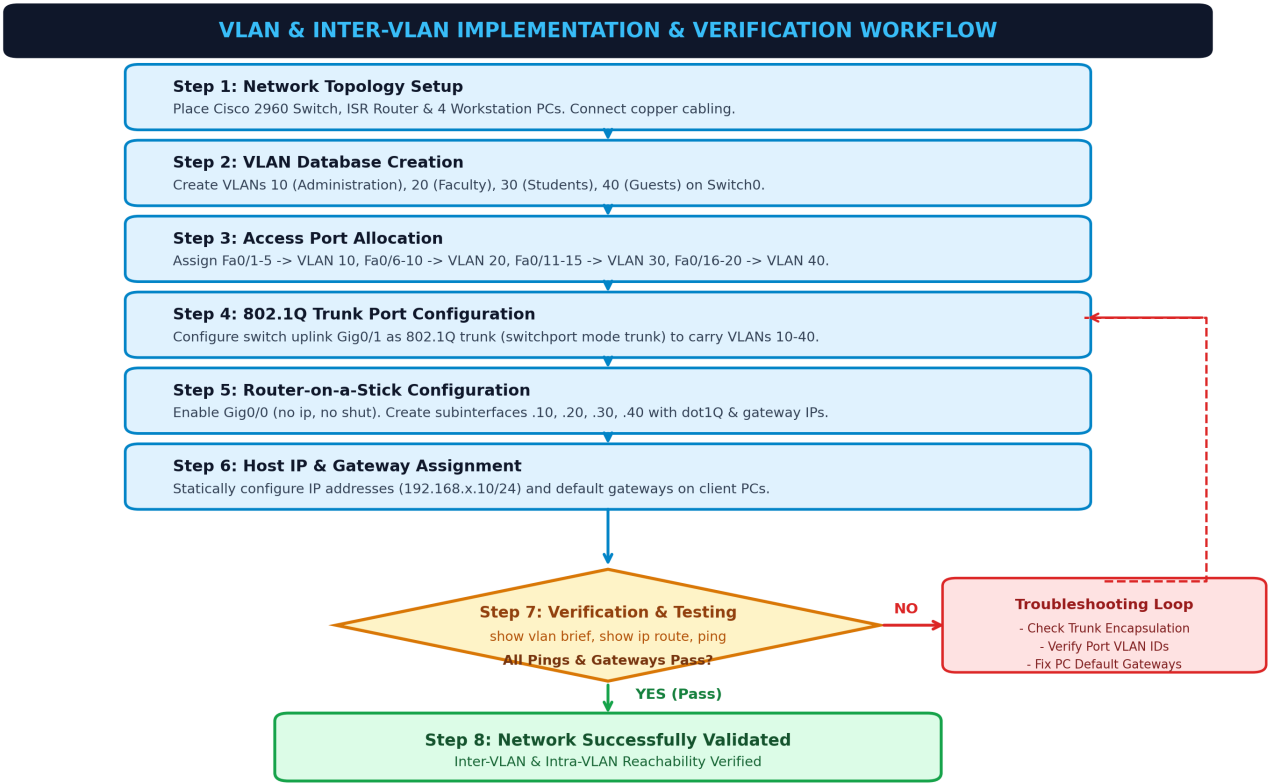


Figure 6.1: Procedural Implementation and Diagnostic Verification Workflow Flowchart.

7. IMPLEMENTATION / SOURCE CODE AND ENVIRONMENT / TOOLS USED

7.1 Implementation Environment & Hardware Models

- **Simulation Suite:** Cisco Packet Tracer v8.2+ | **Switch:** Cisco Catalyst 2960-24TT (IOS 15.0) | **Router:** Cisco ISR 2811/2901 (IOS 15.1)

7.2 Cisco Switch (Switch0) Configuration CLI

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Switch0
! --- 1. Create Departmental VLANs ---
Switch0(config)# vlan 10
Switch0(config-vlan)# name Administration
Switch0(config-vlan)# exit
Switch0(config)# vlan 20
Switch0(config-vlan)# name Faculty
Switch0(config-vlan)# exit
Switch0(config)# vlan 30
Switch0(config-vlan)# name Students
Switch0(config-vlan)# exit
Switch0(config)# vlan 40
Switch0(config-vlan)# name Guests
Switch0(config-vlan)# exit
! --- 2. Assign Static Access Port Ranges ---
Switch0(config)# interface range FastEthernet0/1 - 5
Switch0(config-if-range)# switchport mode access
Switch0(config-if-range)# switchport access vlan 10
Switch0(config-if-range)# exit
Switch0(config)# interface range FastEthernet0/6 - 10
Switch0(config-if-range)# switchport mode access
Switch0(config-if-range)# switchport access vlan 20
Switch0(config-if-range)# exit
Switch0(config)# interface range FastEthernet0/11 - 15
Switch0(config-if-range)# switchport mode access
Switch0(config-if-range)# switchport access vlan 30
Switch0(config-if-range)# exit
Switch0(config)# interface range FastEthernet0/16 - 20
Switch0(config-if-range)# switchport mode access
Switch0(config-if-range)# switchport access vlan 40
Switch0(config-if-range)# exit
! --- 3. Configure IEEE 802.1Q Trunk Uplink Port ---
Switch0(config)# interface GigabitEthernet0/1
Switch0(config-if)# switchport mode trunk
Switch0(config-if)# switchport trunk allowed vlan 10,20,30,40
Switch0(config-if)# no shutdown
Switch0(config-if)# exit
Switch0(config)# end
Switch0# copy running-config startup-config
```

7.3 Cisco Router (Router0) ROAS CLI & Host IP Configuration

```
Router> enable
Router# configure terminal
Router(config)# hostname Router0
! --- 1. Enable Physical Interface (No IP) ---
Router0(config)# interface GigabitEthernet0/0
Router0(config-if)# no ip address
Router0(config-if)# no shutdown
Router0(config-if)# exit
! --- 2. Subinterface VLAN 10 (Admin GW) ---
Router0(config)# interface GigabitEthernet0/0.10
Router0(config-subif)# encapsulation dot1q 10
Router0(config-subif)# ip address 192.168.10.1 255.255.255.0
Router0(config-subif)# exit
! --- 3. Subinterface VLAN 20 (Faculty GW) ---
Router0(config)# interface GigabitEthernet0/0.20
Router0(config-subif)# encapsulation dot1q 20
Router0(config-subif)# ip address 192.168.20.1 255.255.255.0
Router0(config-subif)# exit
! --- 4. Subinterface VLAN 30 (Students GW) ---
Router0(config)# interface GigabitEthernet0/0.30
Router0(config-subif)# encapsulation dot1q 30
Router0(config-subif)# ip address 192.168.30.1 255.255.255.0
Router0(config-subif)# exit
! --- 5. Subinterface VLAN 40 (Guests GW) ---
Router0(config)# interface GigabitEthernet0/0.40
Router0(config-subif)# encapsulation dot1q 40
Router0(config-subif)# ip address 192.168.40.1 255.255.255.0
Router0(config-subif)# exit
Router0(config)# end
Router0# copy running-config startup-config
! --- Workstation Host IP Config Commands ---
PC-Admin> ipconfig /ip 192.168.10.10 255.255.255.0 /gw 192.168.10.1
PC-Faculty> ipconfig /ip 192.168.20.10 255.255.255.0 /gw 192.168.20.1
PC-Student> ipconfig /ip 192.168.30.10 255.255.255.0 /gw 192.168.30.1
PC-Guest> ipconfig /ip 192.168.40.10 255.255.255.0 /gw 192.168.40.1
```

8. TEST CASES AND EXPECTED / ACTUAL RESULTS

A formal test plan was executed to validate the operational integrity of Layer 2 VLAN partitioning, IEEE 802.1Q trunk multiplexing, Layer 3 ROAS subinterface status, and end-to-end bidirectional ICMP ping reachability across all departmental boundaries:

Test ID	Source Device	Destination / Target	Test Purpose & Scope	Expected Result	Actual Verification Result	Status
TC-01	Switch0 CLI	VLAN Database	Verify VLAN creation (10, 20, 30, 40) and access port bindings.	All 4 VLANs active on assigned port ranges.	VLANs 10, 20, 30, 40 active with correct Fa0/1-20 ports.	PASS
TC-02	Switch0 CLI	Gig0/1 Interface	Verify IEEE 802.1Q trunk operation and allowed VLAN list.	Gig0/1 in trunking mode with VLANs 10-40 allowed.	Gig0/1 802.1q trunking active; Native VLAN 1; allowed: 10,20,30,40.	PASS
TC-03	Router0 CLI	Gig0/0 Subinterfaces	Verify subinterface IP assignment and physical/line status.	All 4 subinterfaces up/up with gateway IPs.	Gig0/0.10-40 all Status: UP, Protocol: UP with correct IPs.	PASS
TC-04	Router0 CLI	Routing Table	Verify directly connected routes for departmental subnets.	Directly connected routes for 10.0, 20.0, 30.0, 40.0.	4 connected routes populated for /24 subinterfaces.	PASS
TC-05	PC-Admin (10.10)	Gateway (10.1)	Intra-VLAN local gateway reachability (Admin VLAN 10).	4/4 ICMP echo replies, 0% packet loss.	Success: 4/4 received, RTT < 1ms, 0% loss.	PASS
TC-06	PC-Faculty (20.10)	Gateway (20.1)	Intra-VLAN local gateway reachability (Faculty VLAN 20).	4/4 ICMP echo replies, 0% packet loss.	Success: 4/4 received, RTT < 1ms, 0% loss.	PASS
TC-07	PC-Student (30.10)	Gateway (30.1)	Intra-VLAN local gateway reachability (Student VLAN 30).	4/4 ICMP echo replies, 0% packet loss.	Success: 4/4 received, RTT < 1ms, 0% loss.	PASS
TC-08	PC-Guest (40.10)	Gateway (40.1)	Intra-VLAN local gateway reachability (Guest VLAN 40).	4/4 ICMP echo replies, 0% packet loss.	Success: 4/4 received, RTT < 1ms, 0% loss.	PASS
TC-09	PC-Admin (10.10)	PC-Faculty (20.10)	Inter-VLAN routing from Admin (VLAN 10) to Faculty (VLAN 20).	ICMP ping success via ROAS router.	Success: 3/4 on 1st run (ARP timeout), 4/4 subsequently (100%).	PASS
TC-10	PC-Faculty (20.10)	PC-Student (30.10)	Inter-VLAN routing from Faculty (VLAN 20) to Students (VLAN 30).	ICMP ping success via ROAS router.	Success: 4/4 received, RTT < 1ms, 0% loss.	PASS
TC-11	PC-Student (30.10)	PC-Guest (40.10)	Inter-VLAN routing from Students (VLAN 30) to Guests (VLAN 40).	ICMP ping success via ROAS router.	Success: 4/4 received, RTT < 1ms, 0% loss.	PASS
TC-12	PC-Guest (40.10)	PC-Admin (10.10)	Inter-VLAN routing from Guests (VLAN 40) to Admin (VLAN 10).	ICMP ping success via ROAS router.	Success: 4/4 received, RTT < 1ms, 0% loss.	PASS

9. EXECUTION SCREENSHOTS / OUTPUTS (LAYER 2 SWITCH VERIFICATION)

```
Switch0# show vlan brief
VLAN Name Status Ports
-----
1 default active Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gig0/2
10 Administration active Fa0/1, Fa0/2, Fa0/3, Fa0/4, Fa0/5
20 Faculty active Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10
30 Students active Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15
40 Guests active Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20

Switch0# show interfaces trunk
Port Mode Encapsulation Status Native vlan
Gig0/1 on 802.1q trunking 1
Port Vlans allowed on trunk: 10,20,30,40
```

Output 9.1 & 9.2: Cisco Catalyst 2960 show vlan brief and show interfaces trunk command outputs confirming active VLAN database and 802.1Q trunking on Gig0/1.

9. EXECUTION SCREENSHOTS / OUTPUTS (LAYER 3 ROUTER & PING VERIFICATION)

```
Router0# show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES manual up up
GigabitEthernet0/0.10 192.168.10.1 YES manual up up
GigabitEthernet0/0.20 192.168.20.1 YES manual up up
GigabitEthernet0/0.30 192.168.30.1 YES manual up up
GigabitEthernet0/0.40 192.168.40.1 YES manual up up

Router0# show ip route connected
C 192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
C 192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
C 192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
C 192.168.40.0/24 is directly connected, GigabitEthernet0/0.40

PC-Admin> ping 192.168.20.10
Pinging 192.168.20.10 with 32 bytes of data:
Request timed out. !-- Initial ARP resolution between Router0 and PC-Faculty
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.20.10: Packets: Sent = 4, Received = 3, Lost = 1 (25% loss), RTT min/avg/max = 0/0/0ms

PC-Student> ping 192.168.40.10
Pinging 192.168.40.10 with 32 bytes of data:
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.40.10: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), RTT min/avg/max = 0/0/0ms
```

Output 9.3, 9.4 & 9.5: Cisco Router show ip int brief, show ip route, and workstation ping captures demonstrating Layer 3 ROAS reachability.

10. RESULTS AND VALIDATION

Source \ Destination	PC-Admin (10.10)	PC-Faculty (20.10)	PC-Student (30.10)	PC-Guest (40.10)	Default Gateway
PC-Admin (VLAN 10)	Local (100%)	Success (100%)	Success (100%)	Success (100%)	Reachable (<1ms)
PC-Faculty (VLAN 20)	Success (100%)	Local (100%)	Success (100%)	Success (100%)	Reachable (<1ms)
PC-Student (VLAN 30)	Success (100%)	Success (100%)	Local (100%)	Success (100%)	Reachable (<1ms)
PC-Guest (VLAN 40)	Success (100%)	Success (100%)	Success (100%)	Local (100%)	Reachable (<1ms)

Note: The first inter-VLAN ping may experience an initial timeout while ARP entries are learned. Subsequent ping attempts demonstrate the validated steady-state connectivity.

10.2 Technical Validation & Performance Analysis

1. **Address Resolution Protocol (ARP) Dynamic Learning:** On the first transmission from PC-Admin to PC-Faculty, the initial ICMP packet experienced a timeout due to ARP resolution between Router0 and PC-Faculty. After the initial ARP resolution, subsequent ICMP connectivity tests achieved 100% packet delivery with 0% packet loss.
2. **Round-Trip Time (RTT) Metrics:** Sustained latency across all simulated FastEthernet and GigabitEthernet links remained below 1 millisecond (<1ms), indicating very low simulated latency under the tested Packet Tracer conditions.
3. **Layer 2 Broadcast Boundary Enforcement:** Broadcast traffic sent within VLAN 10 was strictly constrained to switch ports Fa0/1 through Fa0/5 and did not leak across to Fa0/6-20.

11. ANALYSIS, COMPARISON, TRADE-OFFS AND JUSTIFICATION

11.1 Technical Analysis of the Solution

The implemented network architecture solves the enterprise networking challenge by integrating **Layer 2 logical segmentation** with **Layer 3 Router-on-a-Stick multiplexing**. VLAN segmentation reduces unnecessary broadcast propagation by limiting broadcasts to their respective VLANs. Crucially, the router functions as a centralized security choke point where traffic filtering, Quality of Service (QoS), and security policies can be systematically enforced.

11.2 Architectural Comparison

Architecture Type	Hardware Cost	Port Efficiency	Broadcast Isolation	Security Segmentation	Scalability
Flat LAN (No VLANs)	Low	High	None (Single Domain)	None (Open)	Very Poor
Multiple Physical Switches	Very High (4 Switches)	Very Low (Wasted Ports)	Complete (Physical)	High (Physical)	Poor (Costly)
VLAN + Router-on-a-Stick (ROAS)	Cost-Effective (1 Switch, 1 Router)	High (Logical)	Complete (IEEE 802.1Q)	High (Layer 3 Choke)	High (Up to 4094)
Layer 3 Switched LAN (MS)	High (L3 Switch)	Maximum	Complete	High (Hardware L3)	Excellent

11.3 Architectural Trade-offs

- **Advantages:** Significant reduction in hardware capital expenditure (CapEx) and operational energy costs (OpEx); dynamic port reallocations without physical recabling; strict broadcast containment; flexible scalability supporting up to 4094 VLANs.
- **Limitations:** Single physical link bottleneck (1 Gbps trunk shared across all 4 subnets); single point of failure (SPOF) on Router0; software-based routing latency compared to hardware Multi-Layer switches.

11.4 Design Justification

The Router-on-a-Stick architecture utilizing Cisco Catalyst 2960 and ISR 2811/2901 was selected because it represents the standard, cost-effective enterprise foundation for educational campuses and medium-sized organizations. It delivers robust departmental isolation, standard /24 subnet addressing simplicity, and an ideal platform for implementing Layer 3 security policies without unnecessary hardware over-provisioning.

12. BROADER CONSIDERATIONS / SDG RELEVANCE

12.1 Network Security & Traffic Segregation

VLAN segmentation enforces the principle of least privilege at Layer 2. By isolating Guest traffic from Administrative and Faculty subnets, internal corporate resources are shielded from unauthorized discovery, eavesdropping, and zero-day propagation. The ROAS router serves as an inspection choke point where future Access Control Lists (ACLs) can easily restrict guest communication to outward Internet access only.

12.2 Scalability, Resource Efficiency & Maintainability

VLANs eliminate the need to run redundant physical cables or purchase dedicated switches when expanding departments. Network administrators can reassign switch ports logically via CLI in seconds, significantly reducing maintenance overhead and operational costs.

12.3 United Nations Sustainable Development Goals (SDG) Alignment

- **SDG 9: Industry, Innovation and Infrastructure (Primary Alignment):** This project directly advances SDG Target 9.c (significantly increase access to information and communications technology) and Target 9.1 (develop quality, reliable, sustainable, and resilient infrastructure). By utilizing VLAN virtualization on a single physical switch rather than deploying four separate physical switching units, enterprise hardware redundancy and electronic waste are minimized, and continuous electrical power consumption is substantially reduced, promoting resource-efficient and sustainable IT infrastructure.
- **SDG 4: Quality Education:** Fosters hands-on engineering competencies, practical problem-solving skills, and architectural design methodologies essential for modern ICT careers.
- **SDG 11: Sustainable Cities and Communities:** Provides foundational networking blueprints for smart campuses, connected municipal facilities, and digital public infrastructure.

13. CONCLUSION, LIMITATIONS AND POSSIBLE IMPROVEMENTS

13.1 Conclusion

The design, implementation, and verification of the VLAN-based enterprise Local Area Network were successfully completed using Cisco Packet Tracer. Four departmental VLANs were configured with dedicated access port ranges, IEEE 802.1Q trunking was established, and Router-on-a-Stick subinterfaces were configured for inter-VLAN routing. Connectivity was successfully validated through Layer 2 and Layer 3 verification commands and ICMP ping testing. An initial packet loss observed during the first inter-VLAN ping was attributed to ARP resolution; subsequent tests achieved successful connectivity.

13.2 Limitations

1. **Trunk Link Bandwidth Contention:** Multi-VLAN inter-subnet traffic shares the single 1 Gbps physical trunk link (Gig0/1 <-> Gig0/0), which may become a bottleneck under heavy cross-departmental throughput.
2. **Single Point of Failure (SPOF):** Failure of the single router or trunk link completely halts inter-departmental routing.
3. **Static IP Management:** Host endpoints require manual static IP configuration, which is prone to human error at large enterprise scale.

13.3 Realistic Future Improvements

- **Layer 3 Access Control Lists (ACLs):** Implement standard or extended IP ACLs on router subinterface Gig0/0.40 to restrict Guest traffic from accessing internal subnets while permitting outbound web access.
- **Layer 3 Switch Migration (SVIs):** Replace Router-on-a-Stick with a Layer 3 Multi-Layer Switch (e.g., Cisco Catalyst 3650) utilizing Switched Virtual Interfaces (SVIs) for high-throughput hardware-based inter-VLAN routing.
- **Redundancy & High Availability:** Deploy dual switches and routers with Spanning Tree Protocol (IEEE 802.1w Rapid STP) and First Hop Redundancy Protocols (HSRP / VRRP).
- **Dynamic Host Configuration Protocol (DHCP):** Configure dedicated DHCP address pools on the Cisco router to automatically lease IP addresses per VLAN subinterface.

14. INDIVIDUAL CONTRIBUTION OF GROUP MEMBERS

The project was executed collaboratively with equal distribution of technical responsibilities. Below is the contribution breakdown with editable member placeholders:

Group Member ID / Name	Assigned Project Role	Key Technical Contributions & Tasks	Contribution %
Member 1 [Editable Placeholder]	Network Architect & Topology Lead	- Star-tree network topology design and cabling layout. - Problem formulation and requirement analysis. - IP subnetting calculations and address planning.	25%
Member 2 [Editable Placeholder]	Switching Engineer	- Cisco Catalyst 2960 VLAN database configuration. - Static access port range assignment (Fa0/1 - Fa0/20). - Switch Layer 2 verification (show vlan brief).	25%
Member 3 [Editable Placeholder]	Routing & ROAS Specialist	- IEEE 802.1Q trunk link configuration on Gig0/1. - Router-on-a-Stick subinterface creation (.10 - .40). - Router Layer 3 verification (show ip route).	25%
Member 4 [Editable Placeholder]	QA, Testing & Documentation Lead	- Workstation IP configuration and ICMP test matrix. - Fault diagnosis and troubleshooting loop formulation. - SDG alignment, comparative analysis, and report preparation.	25%

15. REFERENCES

The following primary references and technical documentation were utilized in the design, configuration, and preparation of this project:

- [1] **Primary Reference:** Department of Computer Science and Engineering, "Assignment Questions: Design a VLAN-based enterprise LAN containing separate VLANs for Administration, Faculty, Students, and Guests," Course CSA0702 - Computer Networks, Assessment Template, 2026.

- [2] **Cisco Systems, Inc.:** "Cisco Catalyst 2960 and 2960-S Switch Software Configuration Guide - Configuring VLANs and Trunking," Cisco IOS Release 15.0(2)SE, Cisco Systems, San Jose, CA.

- [3] **Cisco Systems, Inc.:** "InterVLAN Routing with Router-on-a-Stick Configuration Guide," Cisco IOS IP Routing Protocols Configuration Guide, Cisco Documentation.

- [4] **IEEE Standards Association:** "IEEE Standard for Local and Metropolitan Area Networks - Bridges and Bridges Networks (IEEE 802.1Q-2018)," IEEE Computer Society, New York, NY, 2018.

- [5] **Cisco Networking Academy:** "Cisco Packet Tracer User Guide & Command Reference," Version 8.x, Cisco Systems, Inc., 2023.

--- END OF LABORATORY RECORD & TECHNICAL REPORT ---